



جامعة المجمعة
Majmaah University

College of Computer and Information Sciences
CS474 — Selected Topics in Data Science

MINI PROJECT

End-to-End Data Science Case Study

WEIGHT 10 Marks	GROUP SIZE 2-3 Members	DURATION 2 Weeks
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Course Information

Academic Year	2025 / 2026
Semester	Spring 2026 (Second Semester)
Course	CS474 — Selected Topics in Data Science
Instructor	Hamad Alabduljabbar
Department	Computer Science (Data Science Track)

Key Dates

Group Enrollments End	Wednesday, April 29, 2026
Dataset Sign-up Closes	Wednesday, April 29, 2026
Submission Deadline	Saturday, May 9, 2026
Presentations	May 11 (Mon) 1:00-3:00 PM & May 12 (Tue) 8:30-10:30 PM (Online)



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1. What is This Project?

This Mini Project is your chance to take everything you've learned in CS474 and apply it to a real-world dataset of your choice. Pick a dataset that interests your group, run a complete data science analysis from start to finish, and tell a clear, compelling story about what you found.

You will work in groups of 2 to 3 students over a 2-week period. By the end, you will have produced a working notebook, a written report, and a short presentation. Together these form a portfolio piece you can show in interviews or on LinkedIn.

Learning Objectives

By completing this project, you will demonstrate that you can:

- Apply the full data science pipeline (problem to insight) on a dataset you have not seen before.
- Use regression and modeling techniques from chapters 3 to 7 of the course on real, messy data.
- Communicate findings clearly to a non-technical audience through visualizations and storytelling.
- Collaborate effectively in a small team and divide work between members.

The Spirit of This Project

This is not a coding contest. We care about the question you ask, the insight you find, and the story you tell. A simple model with a clear interpretation beats a fancy model with no insight.



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2. The 7-Step Pipeline

Your project must work through all 7 steps below. Each step is mandatory, but how deeply you go is your group's choice. Aim for thorough but realistic given your 2-week timeframe.

01 Define	02 Collect	03 Clean	04 Explore	05 Model	06 Evaluate	07 Tell
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01. Problem Definition

State the question you want to answer and identify your target variable Y. Frame why this matters.

Minimum: One paragraph stating your research question, your Y, and who would care about the answer.

Going further: *Connect your question to a real-world stakeholder (e.g. a Riyadh restaurant owner, a car buyer, a city planner).*

02. Data Collection

Document your dataset source, its size, and any sampling or filtering you applied.

Minimum: Source link, original row and column counts, and a brief note if you sampled or filtered the data.

Going further: *Note any data quality concerns or biases you spotted upfront before modeling.*

03. Preprocessing

Clean the data: handle missing values, encode categorical variables, scale features as needed.

Minimum: At least one missing-value strategy, one categorical encoding (from chapter 7), and a clean final dataset.

Going further: *Engineer at least one new feature from existing columns and explain why it might help.*

04. Exploratory Data Analysis (EDA)

Visualize your data to understand distributions, relationships, and surprises.

Minimum: At least 4 meaningful visualizations (histogram, scatter, boxplot, correlation, etc.) with brief written interpretation under each.

Going further: *Document any unexpected pattern that changed how you approached your modeling.*

05. Model Building

Apply at least one regression technique from chapters 3 to 7 to predict your Y.

Minimum: Fit at least one model. Justify your choice in 2 to 3 sentences.

Going further: *Compare two or more models (e.g. simple vs multiple regression, with vs without an interaction term).*



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06. Evaluation

Diagnose whether your model is any good. Check assumptions and report fit metrics.

Minimum: Report at least one fit metric (R-squared or RMSE) and check at least one regression assumption visually.

Going further: *Discuss collinearity (chapter 6) if you used multiple predictors. Identify which features mattered most.*

07. Interpretation & Story

Translate your numbers into insight a non-technical reader can understand and act on.

Minimum: A short conclusion section (in plain English, not statistics jargon) that answers your original question.

Going further: *Suggest one practical recommendation or follow-up question your analysis raises.*



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3. What You Submit

All deliverables are uploaded to Blackboard by the submission deadline. Each group submits one combined ZIP file containing the four items below.

Submission Checklist

- ☐ **Notebook** (Python .ipynb or R .Rmd, your choice). Must run end-to-end without errors.
- ☐ **Report** (PDF, 6 to 10 pages). Use the report template provided on Blackboard.
- ☐ **Slides** (PPTX, 10 to 12 slides). For your online presentation.
- ☐ **Dataset** (CSV file, or a README pointing to the source if the file is too large).

4. Timeline

Two weeks total. Use them deliberately. Suggested pacing below, but feel free to adjust to your group's strengths.

WEEK 1, DAYS 1-3 Setup Pick dataset, sign up, problem definition	WEEK 1, DAYS 4-7 Clean & Explore Preprocessing, EDA, visualizations	WEEK 2, DAYS 8-11 Model & Evaluate Build models, run diagnostics	WEEK 2, DAYS 12-13 Write & Submit Report, slides, final ZIP
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5. Grading: 10 Marks Total

Marks are distributed across four categories. Each category has clear sub-criteria so you know exactly what we are looking for.

EDA 3 marks	MODELING 3 marks	REPORT 2 marks	PRESENTATION 2 marks
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Detailed Rubric

Category	What We Score	Marks
EDA	Data understanding & cleaning quality	1
EDA	Quality and clarity of visualizations	1
EDA	Insights extracted from EDA	1
Modeling	Method choice & justification	1
Modeling	Implementation correctness (notebook runs end-to-end)	1
Modeling	Model evaluation & diagnostics	1
Report	Structure, clarity, and use of the report template	1
Report	Storytelling and quality of insights communicated	1
Presentation	Delivery, flow, and visual quality of slides	1
Presentation	Q&A handling and depth of understanding	1
TOTAL		10



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6. Logistics

Group Formation

Groups must be 2 to 3 students. Form your group via self-enrollment on Blackboard between April 27 and 29. If you are having trouble finding teammates, post on the Blackboard discussion board or speak to the instructor.

Dataset Sign-up

Once your group is formed, choose one dataset from the Datasets Catalog and reserve it on the Google Sheets link below. First-come-first-served. No two groups may pick the same dataset. Sign-up closes on Wednesday, April 29.

[Click here to open the Dataset sign-up sheet](#)

Submission Format

Upload one ZIP file per group through Blackboard. The ZIP must contain your notebook, report (PDF), slides (PPTX), and dataset (or README with link). Name your ZIP after your group number: for example, Group 3 submits Group_3.zip, Group 7 submits Group_7.zip, and so on.

Late Policy

Late submissions lose 10% of the total mark per day, including weekends, up to a maximum of 3 days. After 3 days, late submissions receive zero. Plan your time accordingly: technical issues are not accepted as a reason for late submission.

Presentations

Presentations are ONLINE via Google Meet. Both the meeting link and slot sign-up are inside the Google Sheet linked below. Day 1: Monday, May 11, 2026, from 1:00 to 3:00 PM. Day 2: Tuesday, May 12, 2026, from 8:30 to 10:30 PM. Each group has 15 minutes to present plus 5 minutes for Q&A (20-minute slot total).

[Click here to open the sheet \(meeting link + slot sign-up\)](#)

Where to Get Help

Office hours are open every Sunday and Tuesday after class, or by appointment. Use the Blackboard discussion board for general questions so all groups can benefit from the answer. For dataset-specific or modeling questions, post in the Mini Project channel.

Pick a story worth telling. The data is waiting.